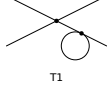


SK Matrix-Model Contributions for All L=1 Topologies

External assignment (by external-propagator appearance): $\phi_r^1(x_1) \phi_a^1(x_2) \phi_r^2(x_3) \phi_a^2(x_4)$, with $x_1 = (t, \mathbf{x})$, $x_2 = (0, \mathbf{0})$, $x_3 = (t, \mathbf{x})$, $x_4 = (0, \mathbf{0})$. Translation table used: same-fold $G_{rr} \rightarrow G_K = \frac{1}{2}(G^> + G^<)$, $G_{ra} \rightarrow G_R$, $G_{ar} \rightarrow G_A$, $G_{aa} \rightarrow 0$; cross-fold $G_{rr}^{12} \rightarrow G^<$, $G_{rr}^{21} \rightarrow G^>$, and any cross-fold propagator with an a endpoint vanishes. Contour ordering convention: contour (2) is later than contour (1), matching `contour_ra_translation.txt`. Each term carries explicit net matrix-size factor N_c^{p-4} (raw loop count p minus the overall $1/N^4$ normalization).

Topology 1 (FeynArts Topology 2)

2 → 2



τ₁

Parsed propagators: 6. Internal vertices: 2. Internal half-edges: 8.

Assignments checked: 1024. Surviving (nonzero vertex weight): 256.

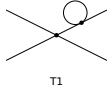
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\mathcal{T}_{1,1} = -16g_2^2 N_c^{-3} G^<(z_1, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_2) G_R(x_1, z_1)$$

$$\mathcal{T}_{1,2} = -16g_2^2 N_c^{-3} G^<(z_1, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_R(x_1, z_1) G_R(z_2, z_2)$$

Topology 2 (FeynArts Topology 2)

2 → 2



τ₁

Parsed propagators: 6. Internal vertices: 2. Internal half-edges: 8.

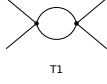
Assignments checked: 1024. Surviving (nonzero vertex weight): 256.

Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 3 (FeynArts Topology 2)

2 → 2

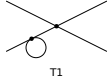


Parsed propagators: 6. Internal vertices: 2. Internal half-edges: 8.
 Assignments checked: 1024. Surviving (nonzero vertex weight): 256.
 Raw N_c power: 0. Net power in $C(t)$: -4. Explicit surviving terms:

$$\mathcal{T}_{3,1} = -16g_2^2 N_c^{-4} G^<(z_1, z_2) G^<(z_1, z_2) G_A(x_2, z_1) G_A(x_4, z_2) G_R(x_1, z_1) G_R(x_3, z_2)$$

Topology 4 (FeynArts Topology 2)

2 → 2



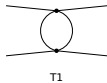
Parsed propagators: 6. Internal vertices: 2. Internal half-edges: 8.
 Assignments checked: 1024. Surviving (nonzero vertex weight): 256.
 Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\mathcal{T}_{4,1} = -16g_2^2 N_c^{-3} G^<(x_1, z_1) G^>(z_1, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_2) G_R(x_3, z_1)$$

$$\mathcal{T}_{4,2} = -16g_2^2 N_c^{-3} G^<(x_1, z_1) G^>(z_1, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_R(x_3, z_1) G_R(z_2, z_2)$$

Topology 5 (FeynArts Topology 2)

2 → 2

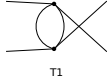


Parsed propagators: 6. Internal vertices: 2. Internal half-edges: 8.
Assignments checked: 1024. Surviving (nonzero vertex weight): 256.
Raw N_c power: 0. Net power in $C(t)$: -4.

No surviving SK contributions for this topology.

Topology 6 (FeynArts Topology 2)

$2 \rightarrow 2$

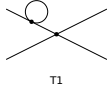


Parsed propagators: 6. Internal vertices: 2. Internal half-edges: 8.
Assignments checked: 1024. Surviving (nonzero vertex weight): 256.
Raw N_c power: 0. Net power in $C(t)$: -4.

No surviving SK contributions for this topology.

Topology 7 (FeynArts Topology 2)

$2 \rightarrow 2$



Parsed propagators: 6. Internal vertices: 2. Internal half-edges: 8.
Assignments checked: 1024. Surviving (nonzero vertex weight): 256.
Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.